



**NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION**

Higher 3

MATHEMATICS

9824/01

Paper 1

27th September 2016

3 Hours

Additional Materials: Answer Paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 7 printed pages.



NANYANG JUNIOR COLLEGE
Internal Examinations

Answer **all** the questions

1 Define $f(x) = x - \ln(1+x)$ where $0 \leq x < 1$.

(i) Prove by differentiation that $f(x)$ is strictly increasing and hence deduce that for $0 < x < 1$,

$$\ln(1+x) < x. \quad [3]$$

(ii) By considering a sequence of transformations that maps the graph of $y = \ln(1+x)$ to the graph of $y = -\ln(1-x)$, or otherwise, show that for $0 < x < 1$,

$$-\ln(1-x) > x. \quad [2]$$

(iii) Deduce from (i) and (ii) that for $n \in \mathbb{C}^+$, $n > 1$,

$$\ln(n+1) - \ln n < \frac{1}{n} < \ln n - \ln(n-1).$$

Hence show that for $N \in \mathbb{C}^+$, $N > 1$,

$$\ln \frac{3N+1}{N} < \sum_{n=N}^{3N} \frac{1}{n} < \ln \frac{3N}{N-1}. \quad [5]$$

(iv) State the exact value of $\lim_{N \rightarrow \infty} \sum_{n=N}^{3N} \frac{1}{n}$. [1]

2 (a) Given that $a > 0$, use the substitution $x = \frac{a}{t}$ to evaluate in terms of a , the integral

$$\int_0^{\infty} \frac{\ln x}{x^2 + a^2} dx. \quad [7]$$

(b) Show that for $n = 0, 1, 2, \dots$,

$$\int_0^{\frac{1}{4}\pi} \tan^{n+2} x dx + \int_0^{\frac{1}{4}\pi} \tan^n x dx = \frac{1}{n+1}. \quad [2]$$

(i) Find the volume of revolution when the region bounded by the curve $y = \tan^2 x$, the x-axis

(ii) Denoting $\int_0^{\frac{1}{4}\pi} \tan^n x dx$ by I_n , prove that for $n > 1$,

$$\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}. \quad [3]$$

Hence deduce the value of the integral $\int_0^{\frac{1}{4}\pi} \tan^{200} x dx$, giving your answer correct to 4 decimal places. [2]

- 3 (i) Prove that for any real numbers x and y ,

$$\frac{1}{2}(x^2 + y^2) \geq xy$$

[1]

- (ii) Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be two finite sequences of positive real numbers.

Denote

$$A_i = \frac{a_i}{\sum_{i=1}^n a_i}, \quad B_i = \frac{b_i}{\sum_{i=1}^n b_i} \quad \text{for } i = 1, 2, \dots, n.$$

Prove that

$$\sum_{i=1}^n A_i B_i \leq 1. \quad [3]$$

Hence prove that

$$\sum_{i=1}^n a_i b_i \leq \sqrt{\sum_{i=1}^n a_i^2 \sum_{i=1}^n b_i^2}, \quad [2]$$

and that equality holds if and only if

$$\frac{a_i}{b_i} = k \quad \text{for all } i = 1, 2, \dots, n,$$

where k is a constant to be determined in terms of $\sum_{i=1}^n a_i^2$ and $\sum_{i=1}^n b_i^2$. [1]

- (iii) In a triangle ABC , use the fact that $\frac{1}{2}(A + B + C) = 90^\circ$ to prove that

$$\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1. \quad [2]$$

By considering two suitable sequences of positive real numbers, deduce from (ii) and the above identity that

$$\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1,$$

justifying your answer.

[3]

- 4 A child has 20 distinct playing cards arranged in a row.
- (a) Show that there is a bijection between the set of possible ways of choosing 6 of these cards so that no two adjacent cards are selected and the set of binary sequences of length 15 and exactly 6 ones, and hence determine the number of different ways of choosing 6 of these cards so that no two adjacent cards are selected. [2]
- (b) Let T_n be the number of ways cards can be chosen from n cards (including the event where no card is chosen) so that no two adjacent cards are selected. Find a recurrence relation for T_n . Hence determine T_6 . [4]
- (c) Eight particular cards are selected.
In how many ways can the cards be distributed
- (i) among 6 friends so that each friend gets at least one card, [3]
- (ii) into 5 piles? [3]

- 5 Pearls of 6 distinct colours are stored in 5 jars. There are 11 pearls of each colour.
- (i) Prove that some jar contains at least 3 pearls of one colour and also at least 3 pearls of another colour. [3]
- (ii) The pearls are then sorted and pearls of the same colour are placed in each jar. Pearls of the remaining colour are then packed into 4 identical bags. Find the number of possible ways of packing the pearls into the 4 bags. [3]
- Assume that the 5 jars now contain red, green, yellow, blue and black pearls, and that there are 11 pearls of the same colour in each jar. Pearls are then chosen from the 5 jars.
- (iii) Find the number of ways of choosing 20 pearls if at least one pearl must be chosen from each jar. [3]
- (iv) Pearls are sewn onto 7 different pieces of jewellery. If only one pearl is sewn onto each piece of jewellery, show that the number of ways a pearl can be sewn onto each of the 7 jewellery pieces, using at least one pearl from each jar, is

$$5^7 - \binom{5}{1} 4^7 + \binom{5}{2} 3^7 - \binom{5}{3} 2^7 + \binom{5}{4} 1^7. \quad [4]$$

6 In this question, $x > 0$.

(i) Show that

(a) $\frac{d}{dx}(x^x) = x^x(1 + \ln x),$ [2]

(b) $\frac{d}{dx}(x^x e^{-x}) = x^x e^{-x} \ln x.$ [1]

A solution to the differential equation

$$\frac{dy}{dx} + y \ln x = x^{-x}$$

has $y = 0$ at $x = 1$.

(ii) Use the Euler method with step size 0.2 to estimate y at $x = 1.4$. [3]

(iii) Copy and complete the table showing the use of the improved Euler method with step size 0.2 to estimate y at $x = 1.4$. [2]

x	y	$x^{-x} - y \ln x$	y^*	$\frac{\Delta y}{\Delta x}$
1	0	1	0.2	$\frac{1+0.767}{2}$
1.2	0.177	0.771	0.331	
1.4				

(iv) Solve the differential equation, expressing your answer in the form $y = f(x)$. Hence find the value of y at $x = 1.4$. [7]

- 7 In a pond, the population P of fish at time t (in weeks) is modelled by the differential equation

$$\frac{dP}{dt} = aP(b - P) - c,$$

where a , b and c are positive constants.

- (i) Explain in context, the significance of the three constants in this model. [3]

It is given that $a = 0.015$ and $b = 200$.

- (ii) Sketch a bifurcation diagram as c varies. Your diagram should consist of five phase lines with corresponding values of c clearly labelled. [4]
- (iii) For the case where $c = 0$, find an expression for P in terms of t given that there are 120 fish in the pond initially. Hence show that in the long run, P will approach a constant value k , where k is to be determined. [5]

- 8 The populations, x and y , of two species of animals at time t days can be modelled by the following system of differential equations

$$\left. \begin{aligned} \frac{dx}{dt} &= 4x - 6y, \\ \frac{dy}{dt} &= 8x - 10y. \end{aligned} \right\} \quad (\mathbf{A})$$

- (i) Show that $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 0$. [3]
- (ii) Hence find the general solution of (A), expressing x and y in terms of t . [4]
- (iii) State what will happen to the populations of the animals in the long run. [1]